Deep Reinforcement Learning for Cryptocurrency Portfolio Management

Comparing Value-Based and Policy Gradient Methods

Jose Márquez Jaramillo, Taylor Hawks

December 2025

Johns Hopkins University
EN.605.741 Reinforcement Learning

Motivation: Why RL for Crypto Portfolios?

The Challenge

- Cryptocurrency markets exhibit **extreme volatility** and regime shifts
- Static allocation rules (equal-weight, market-cap) are suboptimal [1]
- Classical optimization (Mean-Variance) assumes stationarity [6]

Why Reinforcement Learning?

- Focus on **optimal actions**, not predicting returns
- Natural framework for **sequential decision-making**
- Can learn **adaptive, cost-aware** rebalancing policies [3]
- No assumptions about return distributions

Research Questions

1. Can deep RL agents outperform passive baselines?
2. How do value-based vs. policy gradient methods compare?
3. What role does action space design play?

Spoiler: The answers are nuanced!

Project Contributions

Environment Innovations

- **Variable-universe handling**: Dynamic 8–35 assets with canonical indexing (37 assets)
- **Regime-stratified validation**: 5 distinct market conditions
- **Cost-aware reward**: Transaction costs (0.1%) in learning signal
- **Turnover constraints**: 30% limit via projection/penalty

Algorithmic Contributions

- **70-action delta catalog** for discrete RL
- **Dirichlet policy** with GRU encoder

Empirical Contributions

- **7 agents**: 3 baselines + 4 RL methods
- **37 cryptocurrencies** over 7+ years
- **Comparative analysis**: Discrete vs. continuous actions
- **Bayesian hyperparameter search** (Optuna)

Open Source

- Full codebase with reproducible experiments
- Modular environment for future research

MDP Formulation: State & Action Spaces

Formal Definition

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, R, \gamma \rangle$$

- $\mathcal{S}$: States (market data + current holdings)
- $\mathcal{A}$: Actions (target portfolio weights)
- P: Transition dynamics (market evolution)
- R: Reward function (returns − costs)
- γ: Discount factor (future reward weighting)

State Space $\mathcal{S}$

- Market features: $X_t \in \mathbb{R}^{A_t \times 60 \times 4}$
- Previous weights: $w_{t-1} \in \Delta^{A_t}$
- Asset mask: $m_t \in \{0, 1\}^{A_t}$

Action Space $\mathcal{A}$

- Target weights on simplex: $w_t \in \Delta^{A_t}$
- Long-only: $w_{t,i} \geq 0$, $\sum_i w_{t,i} = 1$
- Fully invested (no cash position)

Constraints

- **Turnover limit**: $\|w_t - w_{t-1}\|_1 \leq 0.30$
- **Max allocation**: $w_{t,i} \leq 0.35$
- Enforced via projection (REINFORCE) or penalty (DQN)

MDP Formulation: Reward & Dynamics

Reward Function R

$$r_t = \underbrace{\log(w_{t-1}^\top y_t)}_{\text{portfolio return}} - \underbrace{c\|w_t - w_{t-1}\|_1}_{\text{transaction cost}}$$

- y_t: price relatives ($\text{close}_t / \text{close}_{t-1}$)
- $c = 0.001$ (0.1% transaction fee)
- Log-return: time-additive, geometric growth
- Turnover constraint is **not** in reward—enforced separately

Transition Dynamics P

- Market evolution is exogenous (model-free RL)
- Universe can change (assets enter/exit monthly)
- Previous weights become next state's input

Discount Factor γ

- DQN/DDQN: $\gamma = 0.93$ (via Optuna search)
- High γ enables longer-horizon planning
- REINFORCE: $\gamma = 1.0$ (undiscounted episodic)

Environment Design: Key Innovations

Variable Universe Handling

- Universe size varies: 8–35 assets per day
- Assets enter/exit based on listing dates
- **Canonical indexing**: 37-asset superset
- Padding + masking for consistent NN input

Action Space Implementations

- **DQN/DDQN**: 70 discrete delta actions (adjust top-K, rotate, diversify)
- **REINFORCE**: Continuous Dirichlet sampling

Realistic Constraints

- **Turnover limit**: $\|w_t - w_{t-1}\|_1 \leq 0.30$
- **DQN/DDQN**: Penalty mode ($r = -10$)
- **REINFORCE**: Projection onto feasible set

Feature Engineering

- 4 features per asset: OHLC returns
- 60-day rolling lookback window
- Expanding normalization (no look-ahead)

Cost-Aware Learning

- Costs in reward $\Rightarrow$ learned implicitly
- Not post-hoc adjustment

Agent Architectures: Value-Based vs. Policy Gradient

Value-Based: DQN & DDQN [7, 8]

State Encoding:

- Canonical padding (37 positions)
- Flatten: 8,917 dims $\rightarrow$ 256-dim projection

Q-Network:

- 3-layer MLP: 256$\rightarrow$256$\rightarrow$128$\rightarrow$70
- Experience replay (10K), target net (100 steps)

70-Action Delta Catalog [5]*:*

- Adjust top-K by $\pm$5%, $\pm$10%, $\pm$15%
- Rotate, diversify, concentrate, rebalance

Policy Gradient: REINFORCE [9]

Policy Network:

- Canonical padding (37), no flatten—preserves sequence [2]
- GRU processes full market as single sequence
- **Previous weights** input (turnover awareness)
- 2-layer MLP head $\rightarrow$ per-asset logits

Action Sampling:

- Dirichlet: $\alpha = \text{softplus}(\text{logits})/\tau$
- Temperature τ: tunable exploration (non-standard)

REINFORCE+Baseline:

- Shared GRU encoder with value head
- Advantage: $A_t = G_t - V_\phi(s_t)$

Baseline Strategies

Equal Weight (EW) [1]

- $w_i = 1/N$ for all assets
- Naive diversification
- Minimal turnover
- Strong in volatile markets

Market Cap Weight

- $w_i \propto \text{MarketCap}_i$
- Tracks market portfolio
- BTC/ETH dominated
- Low turnover

Mean-Variance (MVO) [6]

- Rolling 60-day window
- $\max_w \; \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$
- Ledoit-Wolf shrinkage [4]
- Exploits momentum

All baselines use the same environment with identical transaction costs (0.1%) and constraints.

Experimental Setup

Dataset

- **Source**: CoinGecko API
  coingecko.com/en/api
- **Assets**: 37 cryptocurrencies
- **Features**: Daily OHLCV (4 features)
- **Period**: July 2018 – October 2025

Temporal Splits

Split	Period	Days
Warmup	2018-07 – 2018-08	62
Train	2018-09 – 2023-06	1,848
Validation	5 regime windows	$\sim$100
Test	2024-01 – 2025-10	646

Regime-Diverse Validation [3]

- 2018 Crypto Winter
- March 2020 COVID Crash
- 2021 Bull Market
- 2022 Deleveraging
- 2023 Consolidation

Hyperparameter Search

- Optuna with TPE sampler
- 50 trials per agent
- Early stopping + Q-value explosion detection
- Objective: Mean validation return

Results: Performance Summary

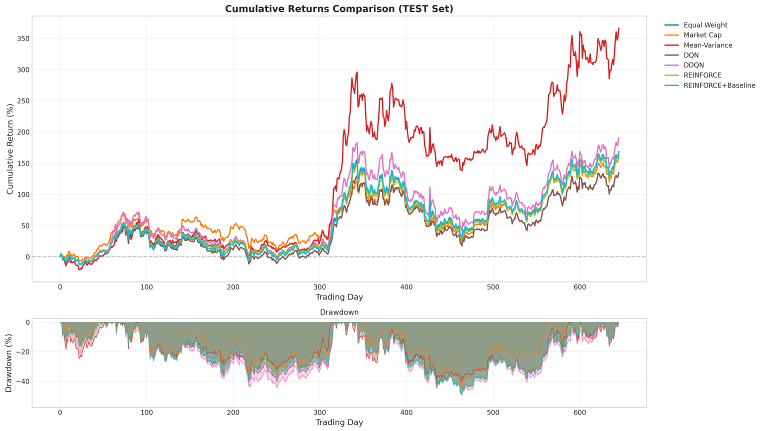
Agent	Ret. (%)	Sharpe	Sortino	MDD (%)	Turn. (%)
Baselines					
Equal Weight	168.5	1.22	1.25	48.9	**0.3**
Market Cap	159.3	1.24	1.30	<u>44.2</u>	0.3
Mean-Variance	**366.5**	**1.64**	**1.69**	**40.0**	13.5
RL Agents					
DQN	135.1	1.09	1.11	49.0	5.7
DDQN	<u>190.8</u>	<u>1.25</u>	<u>1.30</u>	49.9	6.3
REINFORCE	161.8	1.20	1.22	48.8	0.9
REINFORCE+BL	167.4	1.20	1.24	49.3	<u>0.4</u>

Test period: Jan 2024 – Oct 2025 (646 days)
Best in **bold**, second-best <u>underlined</u>

Key Findings

- **MVO dominates**: 366% return, 1.64 Sharpe ratio
- DDQN beats passive baselines: 190.8% vs. 168.5% (EW)
- **REINFORCE competitive**: 167% (w/ baseline) ≈ Equal Weight
- **Risk similar**: All agents experience ∼50% max drawdown

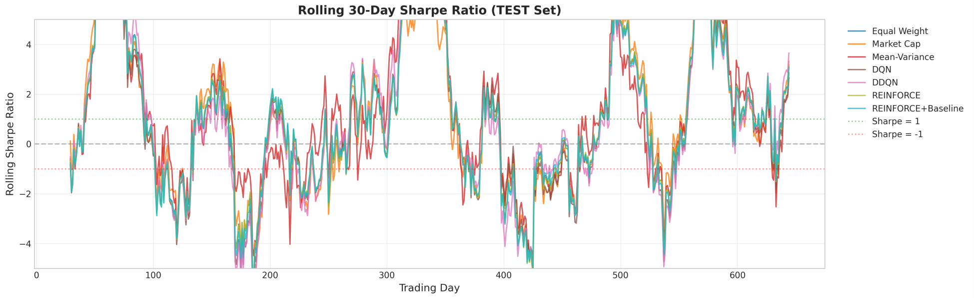
Results: Cumulative Returns & Drawdown



Observations

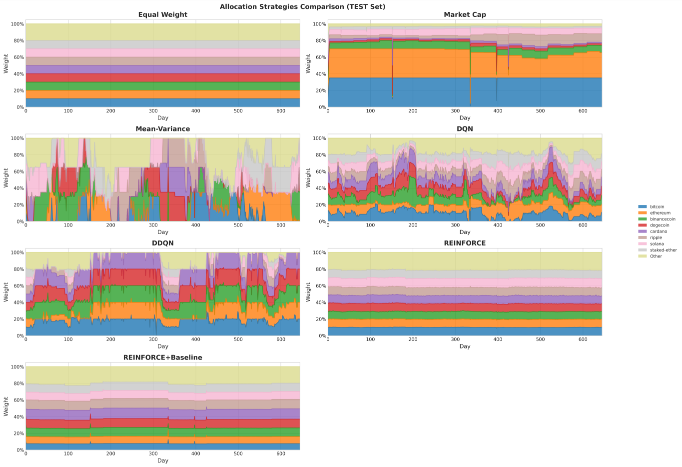
- MVO (red) leads throughout the period
- DDQN (magenta) outperforms passive baselines
- All agents hit ∼50% drawdown mid-2024
- Strong recovery in late 2024
- REINFORCE tracks EW closely

Results: Rolling Risk-Adjusted Performance



MVO maintains highest Sharpe. DDQN and REINFORCE competitive with passive baselines.

Results: Portfolio Allocation Dynamics



EW: stable. Market Cap: BTC/ETH dominated. MVO/DDQN: active rebalancing. REINFORCE: stable allocations under deterministic eval.

Analysis: DDQN vs. REINFORCE Design Tradeoffs

DDQN Success Factors

- **Delta-based actions**: Incremental adjustments with built-in turnover control [5]
- **Penalty mode**: Explicit -10 reward for constraint violations—clear learning signal
- **Double Q-learning**: Reduces overestimation bias [8]
- **Discrete choices**: Easier credit assignment

Key insight: Action space & environment design aligned well for DQN.

REINFORCE Design Considerations

- **No programmed constraints**: REINFORCE has zero hard constraints; any constraint-like behaviour must be learned via environment feedback
- **Continuous outputs**: Unconstrained Dirichlet sampling can propose large jumps
- **Deterministic eval** resolves this: mean $\alpha_i / \sum_j \alpha_j$ produces smooth weights
- **Baseline helped validation, no longer hurt test**: Fixed action selection/sampling bug removed test-time penalty

Lesson: Environment design appropriate for DQN may not suit policy gradient. Output

Analysis: Why Mean-Variance Dominated

MVO Advantages

- **Sample efficient**: Directly estimates μ, Σ from data
- **Closed-form solution**: No gradient descent needed
- **Momentum capture**: Rolling estimates naturally exploit trends
- **Bull market alignment**: Test period favored momentum strategies

Caveats

- **Regime-dependent**: May fail in mean-reverting or crash periods
- **Stationarity assumption**: Violated in crypto markets
- **No adaptation**: Fixed 60-day lookback
- **Higher turnover** (13.5%): More sensitive to estimation noise

Takeaway: Classical methods remain formidable when conditions align!

Limitations

Experimental Limitations

- **Single test regime**: Bull market (2024–2025)
- **Simplified costs**: No slippage or market impact
- **Frozen agents**: No online adaptation post-training
- **Search budget**: 50 trials may miss optimal configs

Design Limitations

- **Deterministic eval**: Stochastic deployment may differ
- **Baseline gap**: Better on validation, narrower on test
- **No cash position**: Fully invested at all times
- **Daily rebalancing**: Weekly might reduce costs
- **Single random seed**: For final training runs

Results should be interpreted as indicative rather than definitive. Multi-regime testing essential for deployment.

Conclusions & Future Directions

Key Takeaways

1. **DDQN beats passive baselines** (190.77% vs. 168.50%)
2. **REINFORCE competitive** with deterministic eval (167%)
3. **Classical methods remain strong**: MVO achieved 366%
4. **Inference behavior matters**: Dirichlet mean vs. sampling

Future Directions

- **Deterministic policies**: DDPG, TD3 as natural middle ground
- **Transformers**: Cross-asset attention for correlation modeling
- **Online learning**: Continuous adaptation to market regimes
- **Multi-regime testing**: Bear markets, crashes, consolidation
- **Stochastic deployment**: Study sampling vs. mean tradeoffs

Code available at: https://github.com/josemarquezjaramillo/crypto-rl-portfolio

Questions?

[1] V. DeMiguel, L. Garlappi, and R. Uppal. Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953, 2009.

[2] Z. Jiang. Cryptocurrency portfolio management with deep reinforcement learning. *arXiv preprint arXiv:1612.01277*, 2016.

[3] Z. Jiang, D. Xu, and J. Liang. A deep reinforcement learning framework for the financial portfolio management problem. *arXiv preprint arXiv:1706.10059*, 2017.

[4] O. Ledoit and M. Wolf. Honey, i shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119, 2004.

[5] G. Lucarelli and M. Borrotti. A deep q-learning portfolio management framework for the cryptocurrency market. *Neural Computing and Applications*, 32:17229–17244, 2020.

[6] H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

[7] V. Mnih, K. Kavukcuoglu, D. Silver, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

References ii

[8] H. van Hasselt, A. Guez, and D. Silver. Deep reinforcement learning with double q-learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 30, 2016.

[9] R. J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3-4):229–256, 1992.